Lab M7.04 – ER-Driven Schema Design (Telecom Domain)

Step 1: Identify Entities

1. Subscribers: this table will store the data related to each subscriber. Attributes would be the subscriber_id (as primary key), the telephone_number and personal data like name and email.
2. Calls: It will gather the data related to the phone call itself. Attributes would be call_id (primary key), caller_id (foreign key of subscriber id), receiver_id (foreign key of subscriber id), call_datetime, call_duration, tower_id (foreign key) and call_type_id (foreign key)
3. Call Types: information related to the type of call. Attributes would be call_type_id (primary key), call_type_name and description.
4. Towers: represents the tower location. Attributes would be tower_id (primary key), tower_name and tower_location.

Step 2: Design Keys & Relationships

<u>Primary keys</u>

1. Subscriber_id: surrogate key (unique, not null and stable).
2. Call_id: surrogate key (unique, not null and stable).
3. Call_type_id: surrogate key (unique, not null and stable).
4. Tower_id: surrogate key (unique, not null and stable).

<u>Foreign keys and realtionships</u>

1. Subscribers: no foreign keys
2. Calls: caller_id and receiver_id are foreign keys of subscribe_id in subscriber table. Call_type_id is a foreign key of call_type_id in Call Type table. Towe_id is a foreign key ot towe_id of Towers table. All relationships are one-to-many
3. Call Types: no foreign keys
4. Towers: no foreign keys

Step 3: Normalize Schema

Subscribers

- subscriber_id (PK)
- telephone_number (unique)
- name
- email (unique)

Calls

- call_id (PK)
- caller_id (FK → subscribers.subscriber_id)
- receiver_id (FK → subscribers.subscriber_id)
- call_datetime

- duration
- call_type_id (FK → call_types.call_type_id)
- tower_id (FK → towers.tower_id

Call_types

- call_type_id (PK)
- call_type_name (unique)
- description

Towers

- tower_id (PK)
- tower_name (unique)
- tower_location

Step 4: ER Diagram Creation

```
+--------------------+
|    SUBSCRIBERS     |
+--------------------+
| PK subscriber_id   |
| telephone_number U |
| name               |
| email U            |
+--------------------+
        | 1
        | makes / receives
        |
        | N
+----------------------------+
|            CALLS           |
+----------------------------+
| PK call_id                 |
| FK caller_id               |
| FK receiver_id             |
| call_datetime              |
| duration                   |
| FK call_type_id            |
| FK tower_id                |
+----------------------------+
        | N              | N
        |                |
        |                |
       1|               1|
+----------------+   +----------------+
|  CALL_TYPES    |   |     TOWERS     |
+----------------+   +----------------+
| PK call_type_id|   | PK tower_id    |
| call_type_name U|  | tower_name U   |
| description    |   | tower_location |
+----------------+   +----------------+
```

Step 5: Schema Justification

Consistency is key as CDR data is used for billing. Accuracy is mandatory. Duplicated data might risk inconsistent billing amounts. Furthermore, with normalization we gain write performance and storge efficiency. 3NF is appropriate in order to remove transit dependency. Changes to one entity don't affect others. Each telephone number is stored once. Joins can reconstruct full information.

Future analytics might be: Total call duration per subscriber, traffic (number of calls) per tower, call type per subscriber.

For analytics, we can embed call type data and tower data with calls in a separate reporting table to avoid repeated joins, pre-aggregate daily call counts